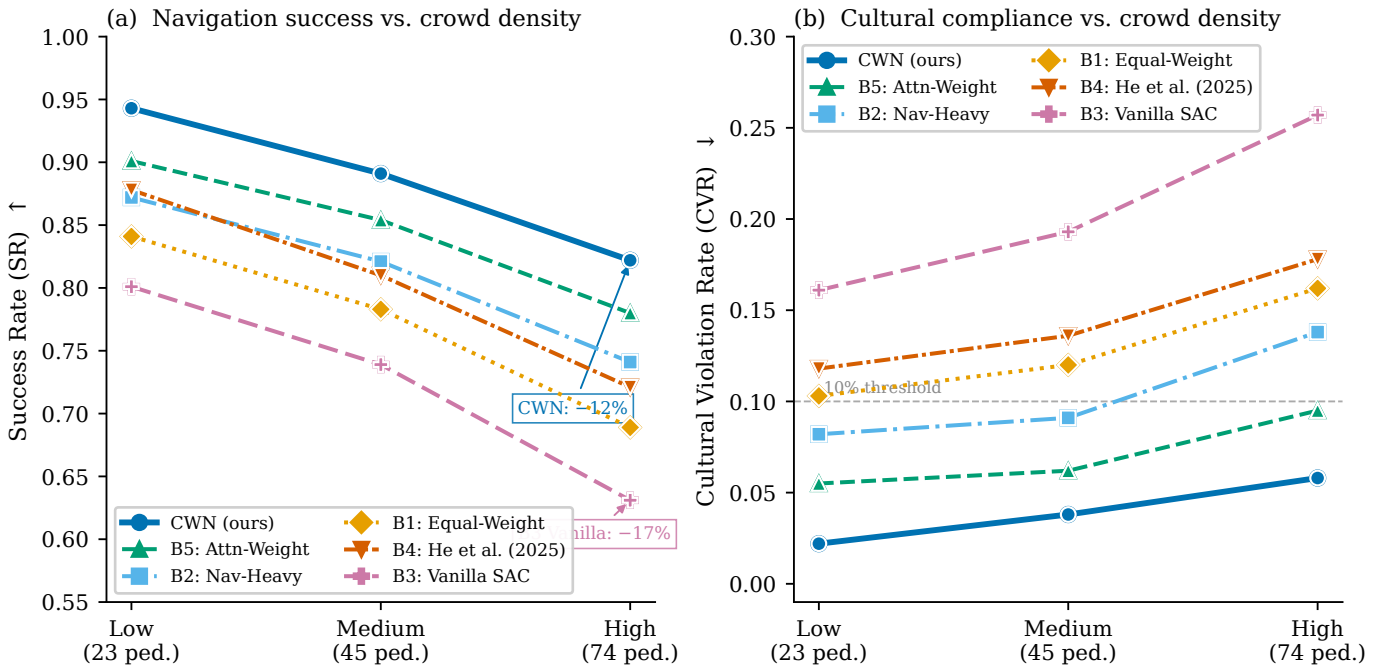
(a) Navigation success vs. crowd density
Success Rate (SR) ↑
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
CWN: −12%
nilla: −17%
CWN (ours)
B5: Attn-Weight
B2: Nav-Heavy
B1: Equal-Weight
B4: He et al. (2025)
B3: Vanilla SAC
Low
(23 ped.)
Medium
(45 ped.)
High
(74 ped.)
(b) Cultural compliance vs. crowd density
Cultural Violation Rate (CVR) ↓
0.30
0.25
0.20
0.15
0.10
0.05
0.00
10% threshold
CWN (ours)
B5: Attn-Weight
B2: Nav-Heavy
B1: Equal-Weight
B4: He et al. (2025)
B3: Vanilla SAC
Low
(23 ped.)
Medium
(45 ped.)
High
(74 ped.)